# Shubhadeep Mandal

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## Education

**International Institute of Information Technology, Hyderabad** July 2025 – Present
*Telangana, India*
*Master of Technology in Computer Science and Engineering* *CGPA: 9.12*

**Indian Institute of Information Technology, Ranchi** Aug 2020 – Jun 2024
*Ranchi, India*
*Bachelor of Technology (Hons.) in Computer Science and Engineering* *CGPA: 9.05*

## Research & Open-Source Experience

**Core Contributor, RookDB** Aug 2026 – Present
*Independent Study, DSAC Lab, IIIT Hyderabad | Rust, SQL, Databases* *Hyderabad, India*

- Co-designed the lab's open-source SQL database in Rust: storage engine, buffer pool, B+Tree indexes and query executor.
- Built the cost-based query planner and `EXPLAIN ANALYZE`, and used them to find and fix slow queries.
- Led performance work that made the benchmark suite **14x faster**; table scans run at **20M rows/s**.
- Main contributor to the engine, with **1,200+ tests** including differential testing against SQLite.

## Projects

**Peer-to-Peer Distributed File Sharing System** | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

**FOOL: Custom POSIX-Style Interactive Shell** | *C++, Linux, POSIX APIs, Signals*

- Built a Unix shell in C++ with pipelines, redirection and job control (background jobs, Ctrl+Z, `fg`/`bg`).
- Fixed race conditions in signal handling so the shell never loses track of finished processes.
- Made startup **52x faster** and added a line editor with tab completion and history.

**Warehouse Traffic Control: Distributed Robot Simulator** | *JavaScript, Three.js, Web Workers*

- Built a 3D simulator of a 200-robot warehouse where all components talk over an unreliable network.
- Designed a lease-based protocol with **zero collisions** over 416K moves, even with lost messages, clock drift and crashes.
- Added replay that rewinds to any moment instantly, and ran benchmark simulations in parallel with Web Workers.

## Technical Skills

**Languages**: C/C++, JavaScript, Rust, Python, SQL, Bash.
**Systems**: Linux, Multithreading, Processes & Signals, Sockets.
**Tools**: Git, Make, ASan/UBSan, strace, Three.js, Docker.

## Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

## Interests

- Reading web novels and manga.
- Playing chess and badminton.